How-To Library

Informatica

Address Verification Best Practices for Russian Federation

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Abstract

This document tells you how to connect the elements in a Russian Federation address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in the Russian Federation.

Supported Versions

- Informatica Address Verification 3.0 and 4.0
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses from the Russian Federation. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Russian Federation address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

An address in the Russian Federation can contain different types of information, including street information and administrative region information. You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Russian Federation addresses:

- BATCH
- COUNTRY RECOGNITION

- FAST COMPLETION
- INTERACTIVE
- PARSE

Character Sets

Russian Federation reference address databases store address data in Cyrillic and Latin scripts. By default, Informatica Address Verification returns the verified Russian Federation addresses in the Cyrillic script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Russian Federation postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: STREET, HOUSE NUMBER[, [BUILDING NUMBER], [SUB-BUILDING NUMBER]]
Line 3: CITY, TOWN, OR VILLAGE
Line 4: [RAYON]
Line 5: [OBLAST, KRAY, OR RESPUBLIKA]
Line 6: POSTAL CODE
```

Note: Rayon, oblast, kray, and respublika are the administrative regions of the Russian Federation.

If the address does not contain Rayon information, the oblast, kray, or respublika information occupies the fourth line of the address and the postal code occupies the fifth line.

If the input address contains Rayon information, Address Verification copies the information to the output address. Use the Locality_2 port for Rayon information.

The following table shows the address lines with sample data from a Russian Federation address:

Information Type	Example
Contact	Ilja Rublijow
Street, House Number	ул. Кирова 94
City, Town, or Village	Геленджик
Oblast, Kray, or Respublika	Краснодарский край
Postal Code	353465

The following sample address illustrates the address in the Cyrillic script:

```
Ilja Rublijow
ул. Кирова 94
Геленджик
Краснодарский край
353465
```

The following sample address illustrates the address in the Latin script:

```
Ilja Rublijow
ul. Kirova 94
Gelendzhik
Krasnodarskiy Kray
353465
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- House Number
- Street
- Province
- Postal Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address data elements to appropriate elements in Address Verification.

The following table shows how you can map the input address data to the address elements:

Address Elements	Address Information	Information Type
Street	ул. Кирова	Street Name
HouseNumber	94	House Number
Locality	Геленджик	City, Town, or Village
Province	Краснодарский край	Administrative Region
PostalCode	353465	Postal Code

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the input address elements within Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
  <AddressElements>
     <Country Type="NAME_EN" Item="1">RUSSIAN FEDERATION</Country>
     <Locality Item="1">Геленджик</Locality>
     <PostalCode Item="1">353465</PostalCode>
     <Province Item="1">Краснодарский край</Province>
     <Street Item="1">ул. Кирова</Street>
     <Number Item="1">94</Number>
  </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.9.0.37682" -->

<Result ProcessStatus="V2"
                ModeUsed="BATCH"
                Count="1"
                CountOverflow="NO"
                CountryISO3="RUS"
                PreferredScript="DATABASE"
                PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
                MailabilityScore="4"
```

```
                ResultPercentage="100.00"
                ElementResultStatus="F0F0E0F04000000000E0"
                ElementInputStatus="60605060200000000060"
                AddressResolutionCode="00000000000000000000"
                ExtElementStatus="00505050100000000000"
                ElementRelevance="10101010100000000010"
                AddressType="S"
                LanguageISO3="RUS">
          <AddressElements>
             <Country Type="NAME_EN" Item="1">RUSSIAN FEDERATION</Country>
             <Locality Item="1">Геленджик</Locality>
             <PostalCode Item="1">353465</PostalCode>
             <Province Item="1">Краснодарский  край</Province>
             <Street Item="1">ул. Кирова</Street>
             <Number Item="1">94</Number>
          </AddressElements>
          <AddressLines>
             <DeliveryAddressLine Line="1">ул. Кирова 94</DeliveryAddressLine>
             <CountrySpecificLocalityLine Line="1">Геленджик</CountrySpecificLocalityLine>
             <FormattedAddressLine Line="2">ул. Кирова 94</FormattedAddressLine>
             <FormattedAddressLine Line="3">Геленджик</FormattedAddressLine>
             <FormattedAddressLine Line="4">Краснодарский  край</FormattedAddressLine>
             <FormattedAddressLine Line="5">353465</FormattedAddressLine>
         </AddressLines>
             <AddressComplete>ул. Кирова 94
             Геленджик
             Краснодарский  край
             353465</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how to verify a Russian Federation address by using SOAP call to Address Verification.

You submit the following request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:v4="http://
validator5.AddressDoctor.com/Webservice5/v4">
   <soapenv:Header/>
   <soapenv:Body>
      <v4:Process>
         <v4:login>*</v4:login>
         <v4:password>*</v4:password>
         <v4:parameters>
          <v4:ProcessMode>BATCH</v4:ProcessMode>
         </v4:parameters>
         <v4:addresses>
          <v4:Address>
             <v4:FormattedAddress>
                 <v4:string>ул. Кирова 94</v4:string>
                 <v4:string>Геленджик</v4:string>
                 <v4:string>Краснодарский край</v4:string>
                 <v4:string>353465</v4:string>
                 <v4:string>RUSSIAN FEDERATION</v4:string>
             </v4:FormattedAddress>
          </v4:Address>
         </v4:addresses>
      </v4:Process>
   </soapenv:Body>
   </soapenv:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
```

```
                  <Result>
                     <ProcessStatus>V2</ProcessStatus>
                     <CountryISO3>RUS</CountryISO3>
                     <ResultDataSet>
                        <ResultData>
                           <ResultNumber>1</ResultNumber>
                           <MailabilityScore>4</MailabilityScore>
                           <ResultPercentage>100.00</ResultPercentage>
                           <ElementInputStatus>6060606020000000060</ElementInputStatus>
                           <ElementResultStatus>F0F0F0F04000000000E0</ElementResultStatus>
                           <ElementRelevance>10101010100000000010</ElementRelevance>
                           <ExtElementStatus>00505050100000000000</ExtElementStatus>
                           <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                           <AddressType>S</AddressType>
                           <LanguageISO3>RUS</LanguageISO3>
                           <Address>
                              <Street>
                                 <string>ул. Кирова</string>
                              </Street>
                              <HouseNumber>
                                 <string>94</string>
                              </HouseNumber>
                              <Locality>
                                 <string>Геленджик</string>
                              </Locality>
                              <PostalCode>
                                 <string>353465</string>
                              </PostalCode>
                              <Province>
                                 <string>Краснодарский край</string>
                              </Province>
                              <Country>
                                 <string>RUSSIAN FEDERATION</string>
                              </Country>
                              <DeliveryAddressLines>
                                 <string>ул. Кирова 94</string>
                              </DeliveryAddressLines>
                              <CountrySpecificLocalityLine>
                                 <string>Геленджик</string>
                              </CountrySpecificLocalityLine>
                              <FormattedAddress>
                                 <string>ул. Кирова 94</string>
                                 <string>Геленджик</string>
                                 <string>Краснодарский край</string>
                                 <string>353465</string>
                                 <string>RUSSIAN FEDERATION</string>
                              </FormattedAddress>
                                 <AddressComplete>ул. Кирова 94;Геленджик;Краснодарский край;
353465;RUSSIAN FEDERATION</AddressComplete>
                           </Address>
                        </ResultData>
                     </ResultDataSet>
                  </Result>
               </Results>
            </ProcessResult>
         </ProcessResponse>
      </soap:Body>
   </soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Russian Federation addresses.

The following table lists the type of enrichments that you can add to a Russian Federation address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments to valid Russian Federation addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Address Enrichments	You can include the Federal Information Addressing System (FIAS) ID in the validated output for Russian Federation addresses. The FIAS ID is an alphanumeric string that identifies the building that contains the address.
Standard Geocoding	You can retrieve the standard, interpolated geocoordinates for valid Russian Federation addresses.

Address Verification Tips and Best Practices

Note the following points when you verify Russian Federation addresses:

- A Russian Federation address can contain information in both Cyrillic and Latin scripts.

  Address Verification returns address information in the Cyrillic script by default. Address Verification can return some address information in the Latin script even if you select the default or Cyrillic script as the preferred script. For example, Address Verification copies building and sub-building information from the input address to the output address and does not transliterate the data from Latin to Cyrillic.

  Address Verification can transliterate all data from Cyrillic to Latin.
- Address Verification does not verify building, sub-building, and rayon information because the reference data does not contain the information.
- Address Verification uses the following transliteration standards to generate an address in the Latin script:

  - BGN/PCGN (1947)

  - ISO 9 (1995)

  If you set the `PreferredScript` attribute to `LATIN`, Address Verification uses the ISO 9 transliteration standard.

  If you set the `PreferredScript` attribute to `LATIN_ALT` or `LATIN_1`, Address Verification uses the BGN/PCGN transliteration standard.
- When you enter a Russian Federation address in a Latin script, do not use English-language terms. The reference address databases for the Russian Federation do not include address data in English. For example, `Lenin Street 4` is street information in English. Address Verification does not verify English-language address data.
- Address Verification assigns oblast, kray, and respublika information to the province element.

Frequently Asked Questions

Does the reference data for the Russian Federation contain district name data?

A Russian Federation address does not require Rayon or district name (Locality_2) information in the final address. The reference data does not contain the information.

If you assign a district name to a Locality element, Address Verification copies the district name from the input to a residue field in the output address.

Do not include a district name in a FormattedAddressLine element in the input address.

Does Address Verification read or write English-language terms in a Russian Federation address?

No. Russian is the official language of the Russian Federation.

English-language street names are growing in popularity in Russia. However, Address Verification reads and writes Russian Federation addresses in the Russian language.

What does a forward-slash symbol indicate between house numbers in a Russian Federation address?

An address in the Russian Federation uses the forward-slash symbol to create a composite house number for a building with addresses on two intersecting streets.

For example, consider the following address:

```
Dobroslobodskaya Ulitsa 7/1 Moscow
```

The address identifies the street and house number as `Dobroslobodskaya Ulitsa 7`. The number `1` identifies the building number on the intersecting street.

How does Address Verification handle rural addresses that do not use street information?

Address Verification copies the locality name to the street field.

For example, consider the following address:

```
Свердловская 1
Свердловская
Свердловская обл
623650
RUS
```

The following example shows the address within AddressElements tags:

```
<AddressElements>
  <Country Type="NAME_EN" Item="1">RUSSIAN FEDERATION</Country>
  <Locality Item="1"> Свердловская</Locality>
  <PostalCode Item="1">623650</PostalCode>
  <Province Item="1"> Свердловская обл</Province>
  <Street Item="1">Свердловская</Street>
  <Number Item="1">1</Number>
</AddressElements>
```

The following example shows the address within AddressLines tags:

```
<AddressLines>
  <FormattedAddressLine Line="1">Свердловская 1</FormattedAddressLine>
  <FormattedAddressLine Line="2">Свердловская</FormattedAddressLine>
  <FormattedAddressLine Line="3">Свердловская обл</FormattedAddressLine>
  <FormattedAddressLine Line="4">623650</FormattedAddressLine>
</AddressLines>
```

The address has the following format on an envelope or label:

```
Свердловская 1
Свердловская
Свердловская обл
623650
RUS
```

Does Address Verification verify house number information?

Yes. Address Verification verifies house number information but does not verify building or sub-building information.

Reference Address Databases

Informatica updates the Russian Federation reference address databases monthly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the UpData newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Russian Federation addresses:

- `RUS5BI.MD`. Batch and interactive modes.
- `RUS5FC.MD`. Fast completion mode.
- `RUS5GC.MD`. Standard geocoding.
- `RUS5E1.MD`. Supplementary data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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